Great 👍 Since this is for **B.E. 5th Semester Computer Networks assignment**, I'll expand the answers with **more depth, detailed explanations, examples, diagrams, and exam-oriented style**.
I'll rewrite each answer in a **descriptive format** so you can directly use it for assignments or exam preparation.

Assignment – Transport Layer (Detailed Answers)

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1. What is the function of Transport Layer?

Definition:
The **Transport Layer** is the **4th layer in the OSI Reference Model** and is responsible for **end-to-end communication** between processes on two hosts. While the network layer delivers packets from host to host, the transport layer ensures that the **right data reaches the right process, correctly, in order, and without duplication.**

Key Functions:

1. **Segmentation and Reassembly:**
   - Splits large messages into smaller segments before transmission.
   - At receiver, reassembles segments into original message.
2. **End-to-End Delivery:**
   - Provides process-to-process delivery (not just host-to-host).
   - Achieved using **port numbers**.
3. **Service Point Addressing (Port Numbers):**
   - Identifies specific processes/applications (e.g., Port 80 → HTTP, Port 25 → SMTP).
4. **Flow Control:**
   - Prevents the sender from overwhelming the receiver's buffer.
5. **Error Control:**
   - Uses acknowledgements (ACKs), checksums, and retransmissions to ensure correctness.
6. **Connection Control:**
   - TCP provides **connection-oriented** service with three-way handshake.
   - UDP provides **connectionless** service.

Example Use Case:

- Downloading a file via HTTP uses **TCP** for reliable transfer.
- Watching a live match stream uses **UDP** for speed over reliability.

Diagram:

```
+-------------------+  Application Layer (HTTP, FTP, DNS)

+-------------------+  Transport Layer (TCP/UDP)

+-------------------+  Network Layer (IP)

+-------------------+  Data Link Layer

+-------------------+  Physical Layer
```

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2. How end-to-end congestion control is provided by TCP?

Definition:
Congestion occurs when **too many packets are present in the network**, causing **packet loss, delays, and reduced throughput**. TCP provides **end-to-end congestion control** by adjusting its sending rate dynamically based on network feedback.

TCP Congestion Control Mechanisms:

1. **Slow Start:**
   - Initial congestion window (**cwnd**) = 1 MSS (Maximum Segment Size).
   - Doubles **cwnd** every Round-Trip Time (RTT).
   - Growth is exponential until **threshold** is reached.
2. **Congestion Avoidance (Additive Increase):**
   - Once threshold is reached, cwnd increases linearly.
   - Prevents sudden overload on network.
3. **Fast Retransmit:**
   - If 3 duplicate ACKs are received, TCP retransmits lost segment **without waiting for timeout**.
4. **Fast Recovery:**
   - Instead of returning to cwnd = 1, TCP reduces cwnd to half and continues linear growth.

Graph (Cwnd vs Time):

```
cwnd

 |            /---- (Linear Growth in Congestion Avoidance)

 |           /

 |   /----/----

 |  / Slow Start (Exponential Growth)

 |__/

  ------------------------------------> Time
```

Conclusion:
TCP avoids congestion by **probing the network capacity** and adjusting cwnd.

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3. Explain TCP segment format and justify the importance of its field values.

TCP Segment Format:

```
 0             15 16            31
+-------------------+-----------------+
| Source Port       | Destination Port |
+-------------------+-----------------+
|               Sequence Number      |
+-------------------------------------+
|           Acknowledgment Number     |
+----+------+--------------------------+
|HLEN| Res. | Flags   |   Window Size   |
+----+------+--------------------------+
|    Checksum       | Urgent Pointer  |
+-------------------------------------+
| Options (if any)    | Data (payload)  |
+-------------------------------------+
```

Field Explanation:

- **Source Port & Destination Port (16 bits):** Identify application processes.
- **Sequence Number (32 bits):** Ensures ordered delivery.
- **Acknowledgment Number (32 bits):** Confirms receipt of data.
- **Header Length (HLEN):** Size of TCP header.
- **Flags (6 bits):**
    - SYN → Connection setup
    - FIN → Connection termination
    - ACK → Acknowledgment
    - RST → Reset connection
    - PSH → Push data immediately
    - URG → Urgent data

- **Window Size:** Controls flow of data (receiver's buffer size).
- **Checksum:** Detects errors.
- **Urgent Pointer:** Points to urgent data.

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4. Difference between Flow Control and Congestion Control

Aspect	Flow Control	Congestion Control
Definition	Ensures sender does not overflow receiver.	Prevents network overload.
Scope	End-to-end (sender and receiver).	Entire network (routers, links, hosts).
Implemented by	TCP sliding window.	TCP congestion algorithms (slow start, etc.).
Goal	Protect receiver buffer.	Maintain network performance.

Example:

- Flow Control = "Don't send more, my buffer is full."
- Congestion Control = "Don't send more, the network is overloaded."

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5. Explain in brief socket, multiplexing and demultiplexing.

Socket:

- Endpoint of communication between two machines.
- Defined by (IP address + Port number).
- Example: 192.168.1.10:80 → HTTP server socket.

Multiplexing (at Sender):

- Multiple applications (e.g., Web, Email, Chat) can send data simultaneously.
- TCP/UDP adds **port numbers** to identify applications.

Demultiplexing (at Receiver):

- Receiver checks port number and delivers data to correct process.

Diagram:

Sender:

App1 → Port 80 ↘

App2 → Port 25 → TCP/UDP → Network

App3 → Port 21 ↗

Receiver:

TCP/UDP → Delivers to App1/App2/App3 using port numbers.

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6. Define error detection and correction.

- **Error Detection:**
  - Identifies corrupted bits during transmission.
  - Techniques:
    - **Parity Bit**
    - **Checksum**
    - **Cyclic Redundancy Check (CRC)**
- **Error Correction:**
  - Corrects errors at the receiver end.
  - Techniques:
    - **Forward Error Correction (FEC)** – Hamming Code.
    - **Automatic Repeat Request (ARQ)** – Stop-and-Wait, Go-Back-N, Selective Repeat.

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7. Explain User Datagram Protocol (UDP) in detail and discuss how it differs from TCP.

UDP Characteristics:

- **Connectionless:** No setup before communication.
- **Unreliable:** No ACKs, no retransmissions.
- **Faster:** Low overhead (8-byte header).
- **Used in:** DNS, Video streaming, VoIP, Online gaming.

UDP Header (8 bytes):

- **Source Port**
- **Destination Port**
- **Length**
- **Checksum**

TCP vs UDP:

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable (ACKs, retransmits)	Unreliable
Ordering	Maintains order	No ordering
Overhead	High (20 bytes header)	Low (8 bytes header)
Speed	Slower	Faster
Use Cases	Web, Email, File Transfer	Streaming, VoIP, DNS

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8. Describe flow control and error control in TCP.

Flow Control in TCP:

- Uses **Sliding Window Protocol**.
- Receiver advertises **window size (rwnd)**.
- Sender limits transmission to available buffer space.

Error Control in TCP:

- **Acknowledgments (ACKs):** Confirm receipt of segments.
- **Checksums:** Detect corrupted segments.
- **Retransmission:** If ACK not received → resend segment.
- **ARQ Techniques:**
  - Stop-and-Wait
  - Go-Back-N
  - Selective Repeat

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9. Describe pseudoheader & Header for checksum calculation in case of UDP.

Checksum Purpose:
Ensures integrity of UDP datagram during transmission.

UDP Pseudoheader (for checksum only):

- Source IP Address
- Destination IP Address
- Protocol Number (17 for UDP)
- UDP Length

UDP Header (8 bytes):

- Source Port
- Destination Port
- Length
- Checksum

Checksum Calculation Process:

1. Sender constructs **Pseudoheader + UDP Header + Data**.
2. Computes checksum and inserts into UDP header.
3. Receiver recomputes checksum → verifies.

✅ These are now **expanded, descriptive, and exam-ready answers** with definitions, key points, tables, diagrams, and comparisons.

👉 Do you want me to also **prepare a neat PDF/Word document with diagrams** (so you can directly submit it as an assignment)?